Responsible AI Framework:
Data & Algorithmic Bias Considerations

NovaCare Health Systems

Course: Responsible Appl of AI (AIML-510-01R4A)

Assignment: 3.4 RAI Framework Project: Data and Algorithmic Bias

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Date: September 12, 2026

NovaCare Health Systems

Fictional healthcare enterprise (consistent with Assignments 1.3 and 1.4)

- **Scale** — 30 hospitals, 150+ outpatient facilities, approximately 45,000 employees
- **Reach** — Serves about 6 million patients annually through hospitals, clinics, diagnostic centers, and telehealth
- **Technology footprint** — Enterprise EHR, cloud analytics, imaging, patient portals, and digital care platforms
- **RAI foundation** — Seven ethical principles established in 1.3, operationalized through the NIST AI RMF (Govern, Map, Measure, Manage) in 1.4

AI Use Cases Under Consideration

Same four use cases established in Assignment 1.3

1

AI-Assisted Medical Imaging

Detects abnormalities in X-rays, CT scans, and MRIs

2

Patient Deterioration Prediction

Predicts patients at risk of rapid clinical decline

3

Generative AI Clinical Assistant

Summarizes records and drafts clinical documentation

4

AI Scheduling & Resource Optimization

Forecasts demand and optimizes staffing and appointments

Chart: Data Sources, Bias Risks, and Mitigation by Use Case

Use Case	Data Source	Potential Bias Source	Mitigation
AI-Assisted Medical Imaging	Historical radiology archive (PACS/EHR)	Underrepresentation of certain skin tones, body types, or anatomies in historical imaging data	*Audit training data for demographic representation; supplement underrepresented groups; validate performance by subgroup*
Patient Deterioration Prediction	EHR vitals, labs, nursing notes	Historical care-access disparities embedded in past treatment and documentation patterns	*Avoid proxy variables tied to past resource allocation; validate outcomes across subgroups; clinical ethics review*
Generative AI Clinical Assistant	Past clinical documentation and provider notes	Inconsistent documentation language/tone across providers for different patient groups	*Human review of every generated note; bias-aware prompt design; clinician feedback loop*
AI Scheduling & Resource Optimization	Historical staffing and scheduling data	Past uneven resource allocation across facility locations	*Regular equity audits by facility/region; objective demand-based rules that do not just replicate history*

Chart: Algorithmic Bias Sources and Mitigations

Organized using the NIST SP 1270 bias framework

Bias Category (NIST SP 1270)	Description	Example at NovaCare	Mitigation
Systemic (institutional/historical)	Bias from institutions and historical inequities, not just data	Deterioration-prediction model shaped by historically unequal access to care	*Equity reviews at the governance level; source data across historically underserved facilities; RAI Steering Committee review*
Statistical/Computational	Errors from unrepresentative samples, proxy variables, or flawed metrics	Imaging model underperforming for underrepresented patient groups	*Subgroup performance testing; pre/in/post-processing bias-mitigation techniques; proxy variable audits*
Human/Cognitive	Bias introduced by developers, labelers, or reviewers during design and oversight	Inconsistent labeling standards used to train the clinical assistant	*Diverse review teams; structured labeling guidelines; ongoing bias-awareness training for staff*

Chart: Testing, Validation, and Monitoring Procedures

Tied to the RAI roles and vetting process defined in Assignment 1.4

Stage	Activity	Frequency / Owner
Pre-Deployment Testing	Subgroup performance testing (accuracy, false positive/negative rates by group); proxy variable audit	*Once, before approval. AI Safety & Validation Lead*
Fairness & Usability Validation	Test outcomes and explanations with real clinicians and patients across diverse groups	*Once, before deployment. Human Factors & Clinical AI Lead*
Independent Validation Gate	Independent review of fairness metrics against a predefined threshold	*Before RAI Approval. AI Risk & Compliance Manager*
Post-Deployment Monitoring	Ongoing tracking of subgroup performance, overrides, and complaints	*Monthly. RAI Governance & Ethics Lead*
Periodic Re-Audit	Full fairness reassessment as patient population or data shifts	*Quarterly/Annual. RAI Steering Committee*

Tools, Processes, and Engagement to Build Trust in Fairness

- Patient and clinician advisory panels for early feedback on new AI tools
- Public-facing model cards summarizing capabilities, limitations, and bias-testing results
- A clear recourse channel for patients/clinicians to report suspected unfair AI outcomes
- Independent, third-party bias audits in addition to internal review
- Ongoing staff training on data ethics and bias recognition
- Periodic transparency reports summarizing fairness-monitoring outcomes

Conclusion

Difficulties and open questions from this workshop

- Deciding what "fair" performance across subgroups even means is harder than it sounds. If two patient populations genuinely have different underlying health profiles, is equal performance the right target, or does that itself distort the model?
- There's a real tension between two of NovaCare's own principles: Fairness & Equity and Patient Safety & Reliability. If fixing a bias issue means giving up some accuracy, that's not a free tradeoff in healthcare the way it might be in hiring or lending, less accuracy could directly cost someone their health.
- Setting a "predefined threshold" for fairness metrics, easy to say and hard to do. Who actually decides what threshold counts as acceptable, and does that threshold apply the same way to every use case?
- Open question: for a use case like AI scheduling, where there's no traditional protected-class lens (race, gender), what does "fairness" even mean, fairness across regions, across staff, across patients?
- Open question: should bias mitigation ever take priority over raw model accuracy when the system directly affects clinical decisions, or is that a line NovaCare shouldn't cross?

References

- Schwartz, R., Vassilev, A., Greene, K., Perine, L., Burt, A., & Hall, P. (2022). Towards a standard for identifying and managing bias in artificial intelligence (NIST Special Publication 1270). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.1270

- Blackman, R. (2022). Explainability: The space between the inputs and the outputs. In Ethical machines: Your concise guide to totally unbiased, transparent, and respectful AI (Ch. 3). Harvard Business Review Press.